Estimating Regional Variation in Intensive Care Bed Requirements in the Coronavirus Pandemic in England: An Observational Trust-Level Comparative Study

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Abstract

Background
The emergence of the COVID-19 pandemic threatens to place unprecedented demand on Critical Care services in the National Health Service in England. An understanding of the comparative accessibility to Critical Care services across England may assist in the focussed expansion of these services nationally during the COVID-19 pandemic.

Methods
Accessibility to Critical Care is calculated for 32,843 Lower Super Output Areas in England based on age stratification and population density adjusted estimates of Critical Care utilisation using the Enhanced Two-Step Floating Catchment Area method. Demand for COVID-19-specific Critical Care is estimated for each trust using 'nearest provider' catchment areas. The number of patients per bed is calculated and the number of additional beds required to reach a ratio of 50 patients per Critical Care bed is quantified.

Findings
Accessibility to Critical Care varies extensively across the country. Accessibility is particularly high in London, Oxford and Cambridge, and is lower in rural areas with long travel distances to reach Critical Care facilities. On average, a three-fold increase in Critical Care capacity would be required to meet a conservative Critical Care capacity of 50 patients per bed. Central urban teaching hospitals and specialist hospitals have particularly low patient to bed ratios, while smaller district general hospitals near to teaching hospitals have particularly high patient to bed ratios.

Interpretation
If the COVID-19 pandemic spreads across England, Critical Care bed availability is well beneath expected demand. Increasing Critical Care capacity is a resource intensive process which must build upon existing infrastructure. Collaborative planning between adjacent hospital trusts to balance relative over and under-provision is required to effectively distribute finite and lifesaving resources across the NHS as a whole.

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Research in context

Evidence before this study
The COVID-19 pandemic threatens to place unprecedented pressure on Critical Care services in the National Health Service in England. While people of all ages may require Critical Care services to treat their COVID-19 infection, it is older patients and those with particular chronic health conditions who are more likely to require ventilatory support. Across England, decades of restructuring of Critical Care services have seen expansion of capacity in teaching hospitals, in comparison to district general hospitals. Recent studies have indicated that the existing Critical Care surge capacity of the NHS is well below expected demand during the COVID-19 pandemic, even after the introduction of a range of mitigation and suppression strategies.

Added value of this study

This study is the first to quantify relative accessibility to Critical Care services in the NHS in England after accounting for age and population density distributions of small geographies, distance to hospitals and the number of available Critical Care beds in each trust. In doing so, variation in the availability of care between rural, suburban and urban contexts is examined, and we find particularly high accessibility in London, Oxford and Cambridge, with lower accessibility in the conurbations of the West Midlands and North West of England. Differences in the ability to meet expected Critical Care demand varies between trusts, with central urban hospitals more able to meet demand due to their higher relative critical care capacity, combined with a younger population less likely to require Critical care.

Implications of all the available evidence

Some parts of England are relatively well served by Critical Care services in comparison to other regions. If the COVID-19 pandemic spreads widely across England, these findings may guide targeted expansion of Critical Care infrastructure in the context of existing capacity. Neighbouring hospital trusts must more readily collaborate to share resources to balance relative over- and under-supply of Critical Care capacity within regions.

Introduction:

The 2020 COVID-19 pandemic is set to create an unprecedented demand on healthcare systems around the world. As of 24th March 2020, the total number of cases globally was 372,757 with 16,231 deaths, of which 8,077 cases have been confirmed in England.[1,2] Of symptomatic cases, between 0.1% of those aged 0-9 years and 27.3% of those aged over 80 years are estimated to require acute inpatient care. Of these cases 5% of those aged 0-9 years to 70.9% of those aged over 80 years are likely to require treatment in an intensive care unit (ICU), with the proportion of both, as well as mortality rates, rising with increasing age.[3] The principal driver of Critical Care admission in COVID-19 patients is respiratory failure requiring invasive or non-invasive ventilatory support. There is an associated requirement for appropriate technical and staffing resources to provide this care.

Epidemiological data from countries with established high levels of community transmission of COVID-19, as well as modelling of expected progression of disease incidence suggest that health system capacity in even well-resourced regions is likely to be insufficient to meet demand, especially where coordinated strategies to mitigate or suppress transmission are not implemented.[3,4] In particular, the demands on Critical Care are estimated to far exceed currently available capacity.[3] These estimates account only for the additional burden of COVID-19 cases on ICUs. It is assumed that there will be no significant change in the rates of emergency Critical Care admission for indications other than COVID-19, even once those cases attributable to elective procedures, which have now largely been suspended in England, are discounted.

Variability in Critical Care capacity in hospital trusts across England, and indeed between the UK and other health systems, pre-dates the current pandemic. Estimates from 2012 are of 6·6 Critical Care beds per 100,000 people in the UK, compared to an OECD average of 11·5 per 100,000, and an estimate of 12·5 per 100,000 in Italy, the OECD country currently most impacted by COVID-19.[5]

Quantifying the need for expansion of Critical Care bed capacity at the level of individual trusts is reliant upon an understanding of the expected demand that a trust will experience coupled with their existing capacity. In the context of contingency planning for a surge in Critical Care demand related to COVID-19, this capacity may include non-standard Critical Care provision. This non-standard provision may include adaptation of other areas with capacity to support invasive ventilation, such as operating theatres, as well as potential conversion or addition of other acute beds to provide level 2 or 3 care. Collectively examining accessibility to these various modes of Critical Care at a national level helps to identify the availability of Critical Care within a reasonable travel tome to hospital. Travel times will assume increasing importance as the burden on public transport and the emergency services increases. Examining the anticipated demand for hospitals in relation to their existing Critical Care capacity can thus help to indicate where specific resources to expand services should be focussed, should the outbreak manifest in their vicinity.

When in need of emergency care, patients are not required to follow prescribed referral patterns, and instead may ignore administrative boundaries and attend the provider of their choice, usually attending the provider nearest to them.[6,7] For emergency presentation of acute respiratory illness, this is anticipated to be the case both for self-presentations, and those via ambulance services. Modelling the potential demand generated by small geographic units of England allows for more granular estimation of the relationship between spatial proximity, demand for care and locally available Critical Care capacity. These three features may be combined as a single accessibility measure in the form of the Enhanced Two-Step Floating Catchment Area (E2SFCA) method. This method is widely used in

healthcare geography to quantify accessibility to clinical services in primary and secondary care.[8–11] E2SFCA estimates local care availability in relation to local demand in a manner which is standardised across all geographic units within a sample, in this case England. This allows for comparison of spatial accessibility to Critical Care across small geographies of England after accounting for variations in local demand, local service capacity and proximity to clinical services.

The ability of health systems to collectively respond to a sudden increase in demand is predicated on the existing availability of clinical capacity and its spatial relationship to clinical need, as well as any changes in such capacity resulting from deployment of additional resources. The spatial spread of the COVID-19 epidemic across England remains to be seen, and will be altered by rapidly changing suppression or mitigation measures to reduce transmission. However, to the extent that the disparate impact of these factors on regional Critical Care utilisation can be used as a basis for determining resource allocation, simulation studies such as that described here may represent one means of optimising the distribution of Critical Care capacity in a health system under exceptional constraints.

Methods:

This study takes publicly available information on the age structure and population densities of small census geographies across England to calculate estimates of local Critical Care demand due to COVID-19. Accessibility to intensive care departments is calculated using the E2SFCA method. Expected numbers of Critical Care attendances are calculated based on defining 'nearest provider' catchment areas for each hospital trust and relating this to their known Critical Care capacity. The expected number of additional Critical Care beds required to reach a level of 50 Critical Care patients per bed is calculated. Based on an expected length of stay of 7 days, 50 Critical Care beds approximates to a full year of bed occupancy by COVID-19 patients, assuming incidence evenly distributed across the year.[12] All results are described and represented spatially using choropleth maps.

Geographic Unit of Aggregation

Smaller geographic units of analysis enable more granular estimation of local variation in estimated demand, and also offer more precise estimates of the time to travel between a patient's home and nearby critical care providers. The Lower Layer Super Output Area (LSOA) is a mutually exclusive, collectively exhaustive census geography, of which there are 32,844 in England, with an average population of around 1,700 individuals. Population-weighted centroids, estimated population age structures and population density values for 2018 are openly available from the Office For National Statistics, and are used to analyse COVID-19 demand and Critical Care availability at a local level.

Calculation of Expected LSOA-level Critical Care Demand

Total number of patients requiring Critical Care admission due to COVID-19 from each LSOA in England was calculated as follows:

Age-specific probabilities of Critical Care admission were calculated by multiplying the probability of hospital admission with the probability of Critical Care admission conditional on being admitted to hospital. Probabilities were taken from Ferguson et al (16th March 2020)[3] and are shown in Table 1:

Age	% of Cases Symptomatic	% of Symptomatic Cases Requiring Hospitalisation	% of Hospitalised Cases Requiring Critical Care	% of Cases Requiring Critical Care
0 to 9	66·7%	0·1%	5·0%	0·0033%
10 to 19	66·7%	0·3%	5·0%	0·010%
20 to 29	66·7%	1·2%	5·0%	0·04%
30 to 39	66·7%	3·2%	5·0%	0·11%
40 to 49	66·7%	4·9%	6·3%	0·21%
50 to 59	66·7%	10·2%	12·2%	0·83%
60 to 69	66·7%	16·6%	27·4%	3·03%
70 to 79	66·7%	24·3%	43·2%	7·00%
80+	66·7%	27·3%	70·9%	12·91%

Table 1: Rates of hospitalisation and Critical Care utilisation by age group. Adapted from Ferguson et al. (2020).

The number of expected cases requiring Critical Care from LSOA (D_k) is calculated in two steps. First, we compute the crude total number of people requiring Critical Care in each LSOA (D_{ck}) if there was 100% transmission by multiplying the number of people in each of the ten-year age bands in Table 1 in each LSOA by the percentage of symptomatic cases requiring critical care. This calculation takes into account the differences in Critical Care requirements due to age and the age structure of each LSOA.

Second, we take into account the influence of population density on the number of cases in a simplified manner. Prior studies have examined the complex relationships between population density and disease transmission, and have often modelled increasing probabilities of transmission with increasing population density using a range of mathematical functions quantifying association between communities or individuals.[13,14] For simplicity, this study will model the probability of transmission as being proportional to the $\log_{10}$ of the population density of individual geographic units of England. Therefore, for each LSOA, the estimated proportion of the population infected (w) was calculated as:

$$w = 0{\cdot}2 \times \log_{10}(\text{density}) \qquad (1)$$

The LSOA with maximum population density according to the Office for National Statistics. has a population density of 106,000 people/km^2.[15] Hence *w* was normalised so that this maximally dense LSOA has 100% of the population infected, whereas a population density of 10,000 people/km^2 is modelled as having 80% of the population infected.

Combining both calculations, we obtain the adjusted number of expected cases requiring Critical Care from each LSOA (D_k) as:

$$D_k = w \times D_{kc} \qquad (2)$$

Travel Time Data

The traffic-free road travel times between all trusts with Critical Care beds and the population-weighted centroids of all LSOAs in England were obtained from https://pythonhealthcare.org/ and supplemented with additional Open Street Map queries (openstreetmap.org) using the

MapBox REST API (mapbox.com). Where trusts were composed of more than one site, the location of the site is the trust with the most inpatient beds determined from the Estates Return Information Classification.[16]

Enhanced Two-Step Floating Catchment Area

Accessibility to Critical Care was calculated for each LSOA in England using the Enhanced Two-Step Floating Catchment Area method (E2SFCA), as follows:[9]

Step 1:

In the first step, the capacity:demand ratio (R_j) is calculated as the ratio of the number of Critical Care beds to the expected demand from the nearby population:

$$R_j = \frac{S_j}{\sum_k^K (D_k \, G(t_{kj}))} \qquad (3)$$

- S_j is the number of Critical Care beds in hospital trust j, derived from NHS Digital in January 2020.[17] This definition includes both ICU and HDU beds.[18]
- D_k is the demand of the location - in this case is the adjusted number of Critical Care patients in LSOA k, of the set of K LSOAs in England.
- $G(t_{kj})$ is the Gaussian distance decay function for the travel time between LSOA k and Hospital j (see below)

A Gaussian decay function was defined to preferentially weight populations closer to a hospital than those further away, as described in Luo and Qi (2009), as follows:

$$G(t_{kj}) = \left(\frac{e^{-0.5\left(\frac{t_{kj}}{t_m}\right)^2} - e^{-0.5}}{1 - e^{-0.5}} \right) \qquad (4)$$

- t_{kj} is the travel time between hospital j and LSOA k
- t_m is the outer limit of the region of LSOAs contributing cases to the hospital j.
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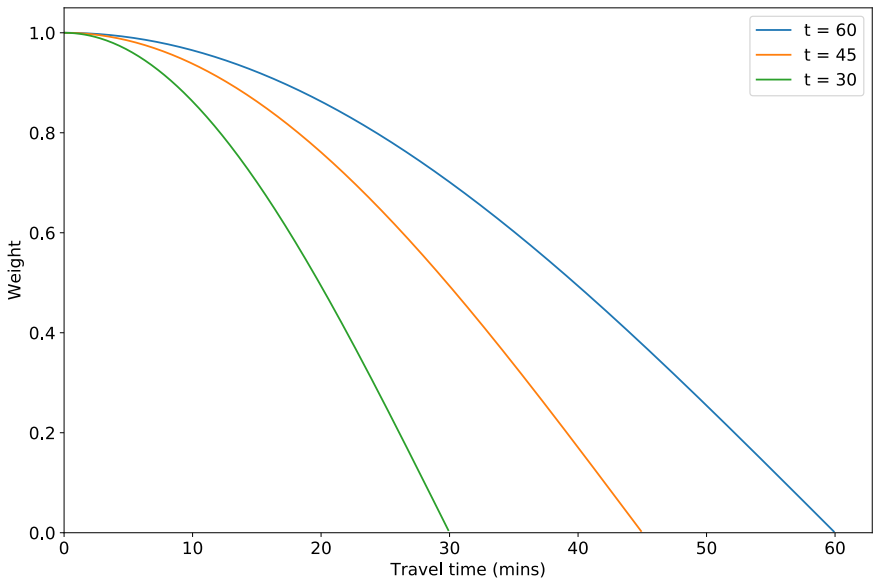

Figure 1: Weights assigned to different Gaussian decay functions over increasing travel time in minutes from an LSOA to a hospital.

Figure 1 shows the weights assigned to travel times with upper limits of t_m = 30, 45 and 60 minutes. In this study, an upper limit (t_m) of 45 minutes travel by road was taken as an appropriate maximal threshold as weights in the range from 0-20 minutes remained above 0·8. Outside of the 45-minute catchment radius, all LSOAs receive a weighting of 0.

Step 2:
In the second step, the accessibility of LSOA k (A_k) is calculated as the sum of the capacity:demand ratio (Rj) of all hospitals within 45 minutes road travel after weighting by the same Gaussian decay function as Equation 4.

$$A_k = \sum_j^J R_j \, G(t_{kj}) \qquad (5)$$

- A_k is the accessibility of Critical Care services for LSOA k
- R_j is the capacity:demand ratio of Hospital j, from the set of J hospital trusts with Critical Care departments.
- $G(t_{kj})$ is the Gaussian distance decay function for the travel time between LSOA k and Hospital j

Estimating Trust-Level Critical Care Capacity Deficits
Nearest provider catchment areas were defined for each hospital trust as the LSOAs for which a trust was the nearest Critical Care department based on road travel time. The overall expected number of Critical Care patients from within this catchment was calculated as the sum of age and population density adjusted demand estimates (D) calculated in Equation 1 for LSOAs within the catchment.

The total number of patients per bed was calculated as the expected number of Critical Care patients in the catchment divided by the total number of Critical Care beds in the trust. A notional rate of 50 patients per bed was taken as a target for expansion of Critical Care capacity. The total number of additional beds required to reach a ratio of 50 patients to each Critical Care bed was calculated for each trust. The geographic distribution of patients to Critical Care beds was mapped, and the number of additional beds required in each catchment was calculated.

Results:
A total of 32,843 LSOAs in England (excluding the Isles of Scilly) were included in the study, representing a total population of 56·0 million individuals. There are two sources of spatial variability in our results. On the one hand, Critical Care demand is influenced by the age structure of the LSOA (as shown in Panel A of Figure 2). The effect of age distribution alone would contribute to rural areas with older populations experiencing greater demand for Critical Care than younger urban populations. On the other hand, population density also accounts for an increased probability of contagion. Population density ranged from 2 people/km^2 to 106,000 people/km^2 with a median of 3,552 people/km^2 for the LSOAs in England (Figure 2, Panel B). Due to this factor, higher probabilities of transmission are encountered in urban areas, although those areas would have lower relative Critical Care demand according to age profiles alone. Panel C of Figure 2 shows the expected numbers of Critical Care presentations after accounting for *both* age and population density.

Age and population-density adjusted estimates of the number of Critical Care admissions from each LSOA ranged from a maximum of 70·8 to a minimum of 2.1, with a median of 17·6 (Figure 2, Panel C). The computations show that demand is highest in suburban areas with

an older population yet relatively high population density. Overall, and estimated 67% of the population are infected over the course of the epidemic based on the crude age and population density adjusted model.

Critical Care Bed Capacity at Baseline

4,113 Critical Care beds were identified across the 136 included trusts in England. The number of beds in each trust ranged from 162 to 5 with a median of 19.[17] For this analysis all available Critical Care capacity was assumed to be allocated to COVID-19 patients.

E2SFCA Accessibility of Critical Care

E2SFCA accessibility values (A_k) ranged from 26 in Trumpington in Cambridge to 0 in the 491 LSOAs that are more than 45 minutes' drive from any Critical Care provider, amounting to a total of 792,961 people. Median accessibility was 6·5. The spatial distribution of accessibility values is shown in Figure 3, Panel A. High areas of accessibility are particularly clustered around London, Cambridge and Oxford, with moderate accessibility close to major towns and cities, and lower accessibility in rural areas, particularly in East Anglia and the South West of England.

Age-group (years)	Number of Critical Care Presentations
0 to 9	157
10 to 19	430
20 to 29	2063
30 to 39	5580
40 to 49	9966
50 to 59	40777
60 to 69	114708
70 to 79	204409
80+	230663

Table 2: Total expected Critical Care presentations after age and population-density weighting, aggregated by age group.

Nearest Provider Catchments Areas

A total of 136 catchment areas were produced consisting of the LSOAs nearest to each hospital trust with a Critical Care department. Catchment populations ranged from 971,971 people to 2,852 people with a median of 389,702. Total patients per bed ranged from 921 to 2 with a median of 239 patients per bed. These stark differences are partly accounted for by the complex and sometimes overlapping multi-site configurations of hospital trusts in London.

13 hospital trusts (10%) had a patient to bed ratio of less than 50. All were teaching hospitals or specialist hospitals, mostly within London. Of those trusts with a patient to bed ratio of more than 50, a median of 63 extra beds per trust is required to reach a ratio of 50 patients per bed. Nationally, to achieve a ratio of 50 patients per bed, a total of 12,174 beds would be required, an increase of 8,061 beds, or a 2·96-fold increase. 48 trusts (35%) would require between a 2 and 5 times increase in Critical Care capacity, 56 trusts (41%) would require between 5 and 10 times increase in Critical Care capacity and 8 trusts (6%) would require a more than ten-fold increase in Critical Care capacity to achieve a ratio of 50 expected patients per bed over the course of the epidemic.

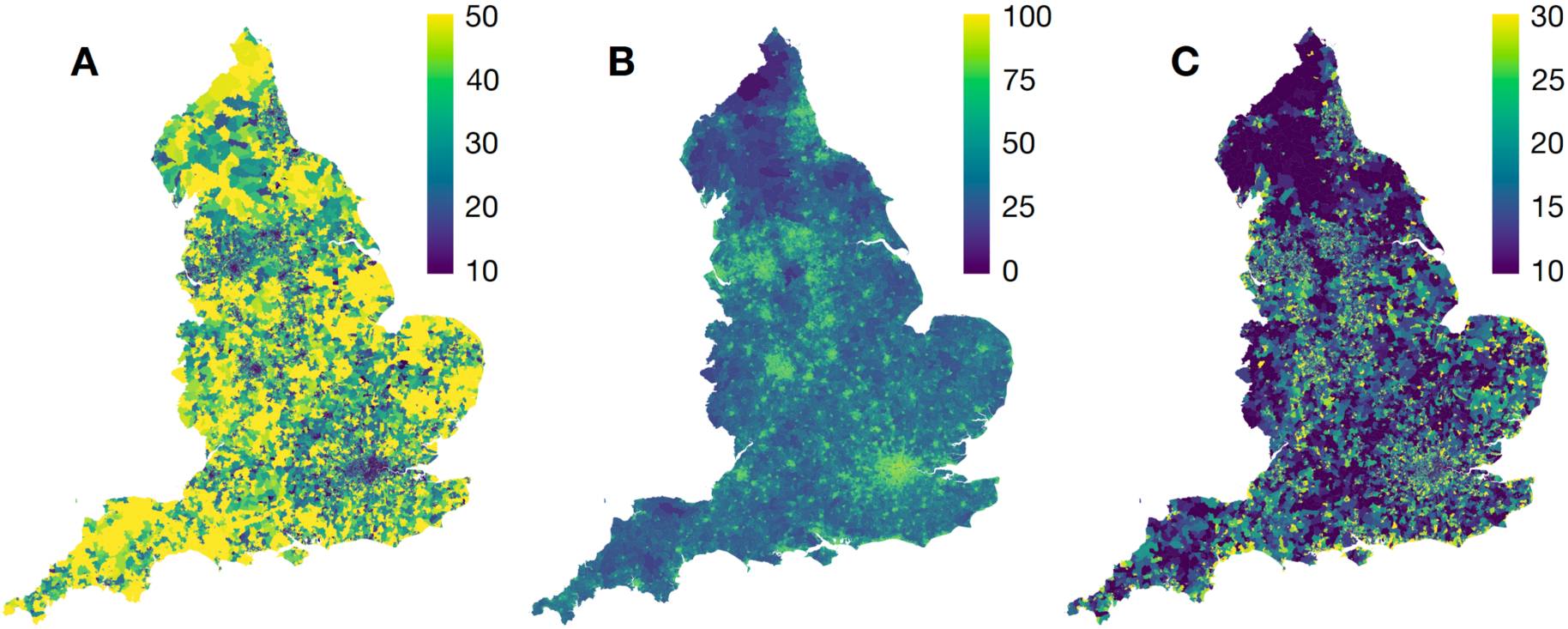

Figure 2: Estimates of Critical Care demand and probability of transmission. **Panel A** shows the estimated number of Critical Care admissions per LSOA after accounting for age structure alone. **Panel B** shows the estimated overall percentage of the population infected based on LSOA-level population density. **Panel C shows** the estimated number of Critical Care admissions per LSOA after accounting for age structure and population density.

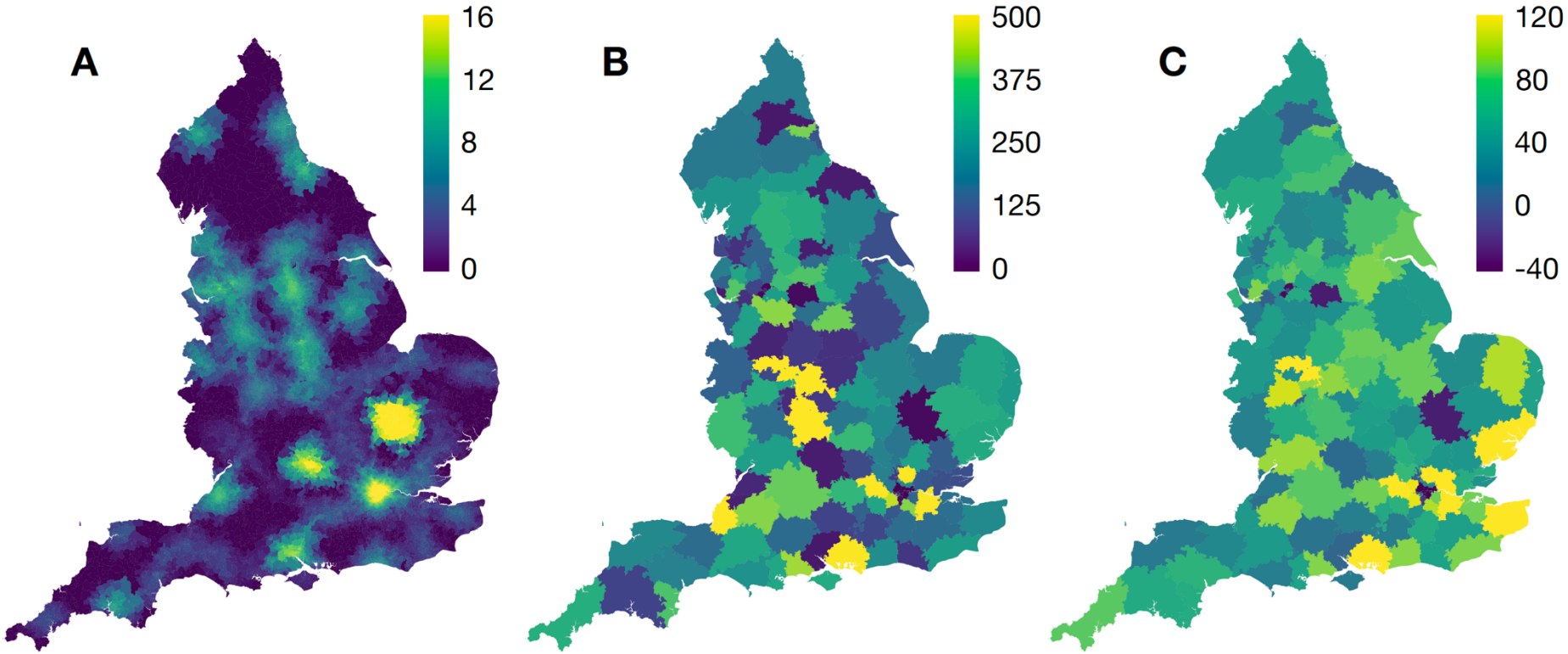

Figure 3: Panel A: E2SFCA estimates of spatial accessibility (A_k) of Critical Care for each LSOA in England based on a t_m of 45 minutes. Higher values indicate greater access to Critical Care. Panel B: Patients per Critical Care bed for each hospital trust 'nearest provider' catchment area. Panel C: Additional beds required to meet a threshold of 50 patients per Critical Care bed in each 'nearest provider' catchment area.

Conclusions and Discussion:
As the COVID-19 pandemic progresses, the healthcare system can expect an unprecedented surge in demand for hospital and critical care beds. With just over four thousand Critical Care beds in England, the number per head of population is relatively low in comparison to other countries.[5] From our modelling, assuming a minimum requirement of 50 patients per Critical Care bed, a median of 68 beds were needed per Trust. This equates to a total of 12,174 Critical Care beds nationally based on the *very conservative estimate* that 50 patients per bed is sufficient. Depending upon the demand experienced during the surge phase of the epidemic, far lower patient to bed ratios will be needed.

Within England, centralisation of services, in teaching hospitals, charged with delivery of an increasing proportion of complex surgical procedures necessitating elective Critical Care admission, has led to disproportionate Critical Care capacity in teaching hospitals compared to the population for whom they are the nearest Critical Care provider. Of note, much of this tertiary-level care is elective, and has been reduced in intensity in anticipation of increased Critical Care demand.[19–21]

Requirements across England varied significantly according to Trust catchment area, with 41% of Trusts needing to increase capacity of Critical Care beds by at least five times, and 6% needing to increase by over ten times. In most cases, the Trusts requiring larger increases in Critical Care capacity are neighboured by Trusts with lower additional Critical Care capacity requirements. Those with lower requirements tend to be teaching hospitals. We found that accessibility is also much higher around these same hospitals; in other words, that those with the least need to expand capacity are those that could more easily expand. Critical Care bed requirements tend to be highest in Trust catchments neighbouring teaching hospitals, indicating the need for strong collaboration between neighbouring Trusts to effectively allocate resources and ensure adequate Critical Care provision.

As of 30th January 2020, an estimated 3,423 (83%) of critical care beds were occupied, of which an estimated 76% are for unplanned care.[17,22] The estimates presented in this analysis therefore represent an conservative estimate of the required additional capacity to provide ongoing Critical Care for emergency patients without COVID-19, even after the cessation of all planned surgical care.

This study contains several simplifying assumptions, simplifications and limitations in our estimation of the local Critical Care demand over the course of the COVID-19 epidemic. First, our modelling estimates - the total number of Critical Care admissions over the whole course of the COVID-19 outbreak - does not attempt to examine the time-dependent, dynamic spread of the condition outbreak across England. Demand is unlikely to be uniformly distributed over time and will fluctuate significantly, and our estimates of required Critical Care capacity therefore do not account for the non-linear rates of disease incidence which, as yet, remain very uncertain. Modelling the probability of disease transmission as proportional to the $\log_{10}$ of population density is also a very simple, and imperfect approach that does not fully embrace the underlying complexities of potential transmission dynamics between populations, particularly in response to changing public health measures to mitigate epidemic spread. More sophisticated, time series transmission models may be applied to this analytical framework in future to model changing local ITU demand to bed ratios over time under a range of alternative mitigation strategies. Additionally, with a focus on Critical Care beds, these models do not account for other potential resource limitations in providing Critical Care. For example, a shortage of ventilators and medical oxygen, as well as lack of adequately trained nursing staff and reduced numbers of medical staff due to infection, may independently limit Critical Care capacity.

The findings also do not account for the duration of a Critical Care stay, an issue which can exacerbate shortages. Previous studies have suggested Critical Care lengths of stay ranging from a median of 7 days in non-survivors to 10 days in survivors.[12,23,24] Our analysis conservatively assumed a 7 day length of stay, but with a length of stay of 10 days, and a requirement of 50 patients per bed, this would result in a demand for occupancy of 500 bed-days over a year which is clearly not feasible. In reality, demand over the course of the outbreak will vary by region and over time, requiring a system which can adapt to local and temporal changes in capacity demands.

The primary aim of this study is to relate the existing Critical Care capacity across England to the potential regional distribution of patients needing ventilatory support in the context of the COVID-19 outbreak. In summary, our modelling predicts very large demand for limited Critical Care capacity nationally, with significant regional and local differences. Our findings may guide the expansion of Critical Care capability and encourage informed collaboration between hospital trusts within the NHS systems as a whole, guided by their relative scarcity of Critical Care capacity.

References:

1. World Health Organisation. *Coronavirus disease 2019 (COVID-19) Situation Report – 57*. (2020).
2. Public Health England. COVID-19 Dashboard. (2020). Available at: https://www.arcgis.com/apps/opsdashboard/index.html#/f94c3c90da5b4e9f9a0b19484dd4bb14. (Accessed: 18th March 2020)
3. Ferguson, N. M. *et al.* Impact of non-pharmaceutical interventions (NPIs) to reduce COVID-19 mortality and healthcare demand. (2020). doi:10.25561/77482
4. Ebrahim, S. H., Ahmed, Q. A., Gozzer, E., Schlagenhauf, P. & Memish, Z. A. Covid-19 and community mitigation strategies in a pandemic. *BMJ* **368**, m1066 (2020).
5. Rhodes, A. *et al.* The variability of critical care bed numbers in Europe. *Intensive Care Med.* **38**, 1647–1653 (2012).
6. NHS. *The NHS Constitution – the NHS belongs to us all*. (2015).
7. Moscelli, G., Siciliani, L., Gutacker, N. & Gravelle, H. Location, quality and choice of hospital: Evidence from England 2002–2013. *Reg. Sci. Urban Econ.* **60**, 112–124 (2016).
8. Wang, F. & Luo, W. Assessing spatial and nonspatial factors for healthcare access: towards an integrated approach to defining health professional shortage areas. *Health Place* **11**, 131–146 (2005).
9. Luo, W. & Qi, Y. An enhanced two-step floating catchment area (E2SFCA) method for measuring spatial accessibility to primary care physicians. *Health Place* **15**, 1100–1107 (2009).
10. Luo, J. *et al.* Use of an E2SFCA Method to Measure and Analyse Spatial Accessibility to Medical Services for Elderly People in Wuhan, China. *Int. J. Environ. Res. Public Health* **15**, 1503 (2018).
11. Bryant, J. & Delamater, P. L. Examination of spatial accessibility at micro- and macro-levels using the enhanced two-step floating catchment area (E2SFCA) method. *Ann. GIS* **25**, 219–229 (2019).
12. Yang, X. *et al.* Clinical course and outcomes of critically ill patients with SARS-CoV-2 pneumonia in Wuhan, China: a single-centered, retrospective, observational study. *Lancet Respir. Med.* (2020). doi:10.1016/S2213-2600(20)30079-5
13. Hu, H., Nigmatulina, K. & Eckhoff, P. The scaling of contact rates with population density for the infectious disease models. *Math. Biosci.* **244**, 125–134 (2013).
14. Truscott, J. & Ferguson, N. M. Evaluating the Adequacy of Gravity Models as a Description of Human Mobility for Epidemic Modelling. (2012).

doi:10.1371/journal.pcbi.1002699
15. Office for National Statistics. Lower layer Super Output Area population density (National Statistics) - Office for National Statistics. (2018). Available at: https://www.ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationestimates/datasets/lowersuperoutputareapopulationdensity. (Accessed: 18th March 2020)
16. NHS Digital. Estates Returns Information Collection - NHS Digital. (2020). Available at: https://digital.nhs.uk/data-and-information/publications/statistical/estates-returns-information-collection. (Accessed: 18th March 2020)
17. NHS Digital. Statistics » Critical Care Bed Capacity and Urgent Operations Cancelled. Available at: https://www.england.nhs.uk/statistics/statistical-work-areas/critical-care-capacity/. (Accessed: 18th March 2020)
18. NHS Digital. *Monthly Trust SitRep (MSitRep) Critical Care Bed Capacity and Urgent Operations Cancelled Definitions and guidance*. (2020).
19. Meadows, C., Rattenberry, W. & Waldmann, C. Centralisation of Specialist Critical Care Services. *J. Intensive Care Soc.* **12**, 87–89 (2011).
20. Bhattarai, N., Mcmeekin, P., Price, C. & Vale, L. Economic evaluations on centralisation of specialised healthcare services: a systematic review of methods. doi:10.1136/bmjopen-2016-011214
21. Imison, C., Sonola, L., Honeyman, M. & Ross, S. *The reconfiguration of clinical services What is the evidence? | The King's Fund*. (2014).
22. Intensive Care National Audit and Research Centre. Public Reporting Website - Home. (2020). Available at: https://onlinereports.icnarc.org/Home. (Accessed: 18th March 2020)
23. Fowler, R. A. *et al.* Critically Ill Patients With Severe Acute Respiratory Syndrome. *JAMA* **290**, 367 (2003).
24. Rello, J. *et al.* Coronavirus Disease 2019 (COVID-19): A critical care perspective beyond China. *Anaesth. Crit. Care Pain Med.* (2020). doi:10.1016/J.ACCPM.2020.03.001

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Authors' contributions:
JMC was involved in the figures, study design, data collection, data analysis and writing. TB, RT, AH and MB were involved in the literature search, data interpretation and writing.

Ethics Statement:
Formal ethical approval was not required for this comparative study using publicly available data.